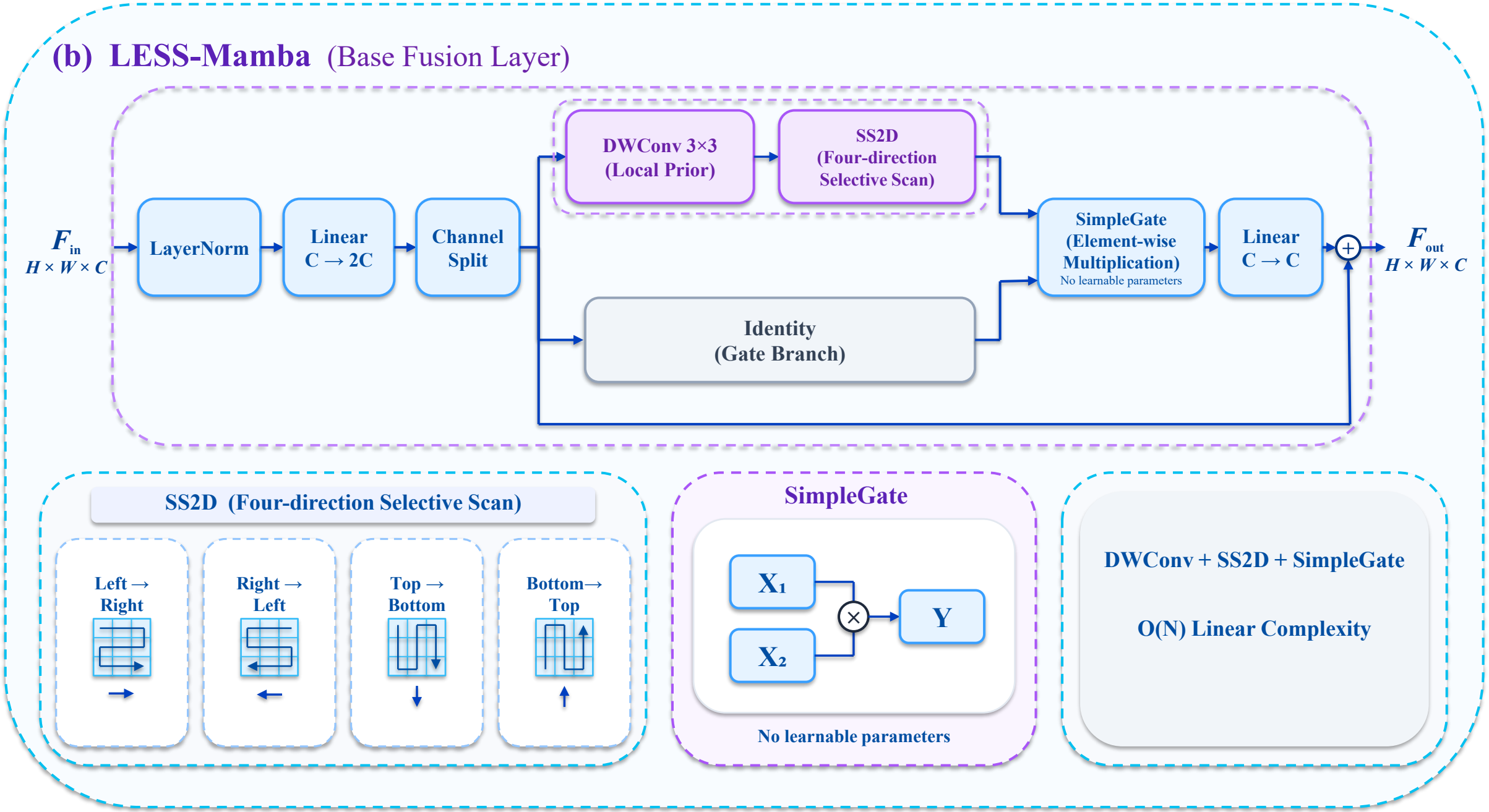

(b) LESS-Mamba (Base Fusion Layer)
F_{in}
$H \times W \times C$
LayerNorm
Linear
C → 2C
Channel
Split
DWConv 3×3
(Local Prior)
SS2D
(Four-direction
Selective Scan)
Identity
(Gate Branch)
SimpleGate
(Element-wise
Multiplication)
No learnable parameters
Linear
C → C
F_{out}
$H \times W \times C$
SS2D (Four-direction Selective Scan)
Left →
Right
Right →
Left
Top →
Bottom
Bottom→
Top
SimpleGate
X1
X2
Y
No learnable parameters
DWConv + SS2D + SimpleGate
O(N) Linear Complexity